

JINGMIN SUN

Software Machine Learning Engineer, Ph.D.

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EMPLOYMENT

Software Machine Learning Engineer
Teradyne, Inc.

- AI Enablement Team

Sep 2026 – Present
North Reading, MA

Postdoctoral Fellow
Johns Hopkins University

- Department of Applied Mathematics and Statistics
- Advisor: Prof. Mauro Maggioni

Jul 2025 – Sep 2026
Baltimore, MD

EDUCATION

Ph.D. in Mathematical Science
Carnegie Mellon University

- Advisor: Prof. Hayden Schaeffer (UCLA)
- Research area: Operator learning, PDE foundation models, Optimization, Control problems

Sep 2020 – May 2025
Pittsburgh, PA

B.S. in Mathematics
Rensselaer Polytechnic Institute

- GPA: 3.98/4.00; *Summa Cum Laude*

Sep 2017 – Dec 2019
Troy, NY

PUBLICATIONS

1. Derek Jollie, **Jingmin Sun**, Zecheng Zhang, and Hayden Schaeffer. Time-series forecasting and refinement within a multimodal PDE foundation model. *Journal of Machine Learning for Modeling and Computing*, 6(2), 77–89.
2. **Jingmin Sun**, Yuxuan Liu, Zecheng Zhang, and Hayden Schaeffer. Towards a foundation model for partial differential equations: Multi-operator learning and extrapolation. *Physical Review E*, 111(3), 035304. <https://doi.org/10.1103/PhysRevE.111.035304>
3. Yuxuan Liu, **Jingmin Sun**, Xinjie He, Griffin Pinney, Zecheng Zhang, and Hayden Schaeffer. PROSE-FD: A Multimodal PDE Foundation Model for Learning Multiple Operators for Forecasting Fluid Dynamics. arXiv:2409.09811 (2024). *Foundation Models for Science Workshop at NeurIPS 2024*.

PREPRINTS

1. **Jingmin Sun**, Zecheng Zhang, and Hayden Schaeffer. LeMON: Learning to Learn Multi-Operator Networks. arXiv:2408.16168 (2024). Under review at *Neurocomputing*.
2. Yuxuan Liu, **Jingmin Sun**, and Hayden Schaeffer. BCAT: A Block Causal Transformer for PDE Foundation Models for Fluid Dynamics. arXiv:2501.18972 (2025). Under review at *Journal of Artificial Intelligence Research*.
3. Adrien Weihs, **Jingmin Sun**, Zecheng Zhang, and Hayden Schaeffer. A Deep Learning Framework for Multi-Operator Learning: Architectures and Approximation Theory. arXiv:2510.25379 (2025). Under review.
4. Min Zhu, **Jingmin Sun**, Zecheng Zhang, Hayden Schaeffer, and Lu Lu. PI-MFM: Physics-informed multimodal foundation model for solving partial differential equations. arXiv:2512.23056 (2025). Under review.

PAPERS IN PREPARATION

1. **Jingmin Sun** and Mauro Maggioni. On machine learning models for cardiac fibrosis.
2. **Jingmin Sun**, Yantao Wu, and Mauro Maggioni. On stability of scientific machine learning models.

PROJECTS

Enhancing the Interpretability of the PDE Foundation Model Jun 2024 – Present

- With Prof. Lu Lu's group, Prof. Zecheng Zhang, and Prof. Hayden Schaeffer

Symbolic Information Analysis for PDE Foundation Model Jun 2024 – Present

- With Prof. Giang Tran's group, Prof. Hans De Sterck, and Prof. Hayden Schaeffer

Efficiency and Memory Enhancement of PDE Foundation Model Jun 2024 – Present

- With Prof. Zecheng Zhang and Prof. Hayden Schaeffer's group

BelNet on Dynamical Systems May 2023 – Present

- With Prof. Zecheng Zhang and Prof. Hayden Schaeffer

Kernel Analog Forecasting with Controls May 2022 – May 2023

- With Prof. Rachel Ward and Prof. Hayden Schaeffer

On Sticky Brownian Motion and Numerical Solution Jan 2020 – May 2020

- Rensselaer Polytechnic Institute, with Prof. Fengyan Li

TEACHING & MENTORING

Instructor

EN.553.481/681 Numerical Analysis

Johns Hopkins University

Spring 2026

REU Co-Mentor

Research Experience for Undergraduates

University of California, Los Angeles

Summer 2024

Teaching Assistant

Numerical Methods

Carnegie Mellon University

Spring, Fall 2024

Computational Linear Algebra

Fall 2023

Linear Algebra for Data Science

Fall 2023

Probability (Graduate Level)

Spring 2023

Numerical PDEs

Fall 2022

Integration and Approximation

Spring 2022

Probability

Fall 2021

Matrix Algebra

Summer 2021

Numerical Linear Algebra

Spring 2021

Principles of Analysis I

Fall 2020

Teaching Assistant

Linear Algebra

Rensselaer Polytechnic Institute

Spring 2020

AWARDS

2024 Travel Award, SIAM Conference on Mathematics of Data Science

2020 The Max Hirsch Prize, Rensselaer Polytechnic Institute

PRESENTATIONS

PDE Foundation Model: Generalization, Learning to Learn and More <i>AI and PDE Workshop @ ICLR 2026</i>	Apr 2026
PDE Foundation Model: Generalization, Learning to Learn and More <i>SIAM Conference on Uncertainty Quantification</i>	Mar 2026
PDE Foundation Model: Generalization, Learning to Learn and More <i>SIAM New York-New Jersey-Pennsylvania Section Conference</i>	Nov 2025
PDE Foundation Model: Generalization, Learning to Learn and More <i>Data Science Seminar @ Johns Hopkins University</i>	Sep 2025
Towards a Foundation Model for PDEs: Multi-Operator Learning and Extrapolation <i>SIAM Conference on Mathematics of Data Science</i>	Oct 2024
LeMON: Learning to Learn Multi-Operator Networks <i>Prof. Lu Lu's Seminar, Yale University</i>	Oct 2024
PDE Foundation Model: Generalization, Meta-learning and More <i>Applied and Computational Math Seminar, Florida State University</i>	Sep 2024
Predicting Operators and Symbolic Expressions using Multimodal Transformers – PDE <i>Prof. Hayden Schaeffer's Seminar, University of California, Los Angeles</i>	Mar 2024

SKILLS

Machine Learning Foundation model engineering (e.g., transfer learning, meta-learning);
Python (NumPy, PyTorch, JAX, TensorFlow, SciPy, scikit-learn)

Computing High-performance computing (Linux/Unix, Bash, CUDA); MATLAB; R

Tools $\text{\LaTeX}$, Markdown, Git/GitHub, HTML/CSS

Languages English, Chinese (Native)